Systems Analysis and Control

Matthew M. Peet
Arizona State University

Lecture 6: Calculating the Transfer Function

Introduction

In this Lecture, you will learn: Transfer Functions

- Transfer Function Representation of a System
- State-Space to Transfer Function
- Direct Calculation of Transfer Functions

Block Diagram Algebra

- Modeling in the Frequency Domain
- Reducing Block Diagrams

Previously:

The Laplace Transform of a Signal

Definition: We defined the Laplace transform of a **Signal**.

- **Input**, $\hat{u} = \mathcal{L}(u)$.
- **Output**, $\hat{y} = \mathcal{L}(y)$

Theorem 1.

Any bounded, linear, causal, time-invariant system, G, has a **Transfer Function**, $\hat{G}$, *so that if $y = Gu$, then*

$$\hat{y}(s) = \hat{G}(s)\hat{u}(s)$$

There are several ways of finding the *Transfer Function*.

The Laplace Transform of a State-Space System

Transfer Functions

Recall the State-Space System

$$\begin{aligned} \dot{x} &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \qquad x(0) = 0 \end{aligned}$$

Applying the Laplace Transform to the first signal:

$$s\hat{x}(s) - x(0) = A\hat{x}(s) + B\hat{u}(s)$$

where we used

- Linearity
- Differentiation Property

Since $x(0) = 0$ we have

$$(sI - A)\hat{x}(s) = B\hat{u}(s) \qquad \text{or} \qquad \hat{x}(s) = (sI - A)^{-1}B\hat{u}(s)$$

The Laplace Transform of a State-Space System

Transfer Functions

Applying the Laplace Transform to the output equation,

$$y(t) = Cx(t) + Du(t)$$

we get

$$\hat{y}(s) = C\hat{x}(s) + D\hat{u}(s)$$

Solving for $\hat{y}(s)$, we get

$$\hat{y}(s) = \left(C(sI - A)^{-1}B + D\right)\hat{u}(s)$$

Thus we have a Transfer Function for the system which can be used to construct a solution for any input using the frequency-domain representation.

Theorem 2.

The **Transfer Function** *for a state-space system is*

$$\hat{G}(s) = C(sI - A)^{-1}B + D$$

Question: How to find $(sI - A)^{-1}$?

Applying the Laplace Transform to the output equation,

$$y(t) = Cx(t) + Du(t)$$

we get

$$\hat{y}(s) = C\hat{x}(s) + D\hat{u}(s)$$

Solving for $\hat{y}(s)$, we get

$$\hat{y}(s) = \left(C(sI - A)^{-1}B + D\right)\hat{u}(s)$$

Thus we have a Transfer Function for the system which can be used to construct a solution for any input using the frequency-domain representation.

Theorem 2.

The **Transfer Function** *for a state-space system is*

$$\hat{G}(s) = C(sI - A)^{-1}B + D$$

Question: How to find $(sI - A)^{-1}$?

Technically, you could use

$$(sI - A)^{-1} = \frac{\text{adj}(sI - A)}{\det(sI - A)}$$

where

$$\text{adj}(M) = [C_{ji}] \qquad C_{ij} = (-1)^{i+j}\det(M_{ij})$$

where M_{ij} is the submatrix obtained by removing row i and column j.

In MATLAB just use the commands: `sys=ss(A,B,C,D)` and `G=ss2tf(sys)`.

Finding Transfer Functions from Dynamics

The General Approach

1. Write down the differential equations
   1.1 including input $u(t)$, output $y(t)$, and any states, $x(t)$
2. Apply the Laplace transform to all equations.
   2.1 Resulting equations are in $\hat{u}(s)$, $\hat{y}(s)$, and $\hat{x}(s)$.
   2.2 If there are n states, $\hat{x}(s)$, you should have $n+1$ equations (one for $\hat{y}(s)$).
3. Solve for $\hat{x}(s)$ in terms of $\hat{u}(s)$
4. Solve for output $\hat{y}(s)$ in terms of input $\hat{u}(s)$
5. Then the transfer function is $\hat{G}(s) := \frac{\hat{y}(s)}{\hat{u}(s)}$

Finding Transfer Functions from Dynamics

Example: Simple State-Space System

Step 1: Write down the differential equations

$$\dot{x}(t) = -x(t) + u(t)$$
$$y(t) = x(t) - .5u(t) \qquad x(0) = 0$$

Step 2: Apply the Laplace transform to all equations:

$$\mathcal{L}\big(\dot{x}(t) = -x(t) + u(t)\big) \quad \Rightarrow \quad s\hat{x}(s) = -\hat{x}(s) + \hat{u}(s)$$
$$\mathcal{L}\big(y(t) = x(t) - .5u(t)\big) \quad \Rightarrow \quad \hat{y}(s) = \hat{x}(s) - .5\hat{u}(s)$$

Step 3: Solve for $\hat{x}(s)$ in terms of $\hat{u}(s)$

$$(s+1)\hat{x}(s) = \hat{u}(s) \qquad \text{and so} \qquad \hat{x}(s) = \frac{1}{s+1}\hat{u}(s).$$

Step 4: Solve for output $\hat{y}(s)$ in terms of input $\hat{u}(s)$

$$\hat{y}(s) = \hat{x}(s) - .5\hat{u}(s) = \frac{1}{s+1}\hat{u}(s) - .5\hat{u}(s) = \frac{1 - .5(s+1)}{s+1}\hat{u}(s) = \frac{1}{2}\frac{s-1}{s+1}\hat{u}(s)$$

Step 5: Thus we have the Transfer Function:

$$\hat{G}(s) = \frac{\hat{y}(s)}{\hat{u}(s)} = \frac{1}{2}\frac{s-1}{s+1}$$

Using a Transfer Function to Find the Output

Example: Step Input

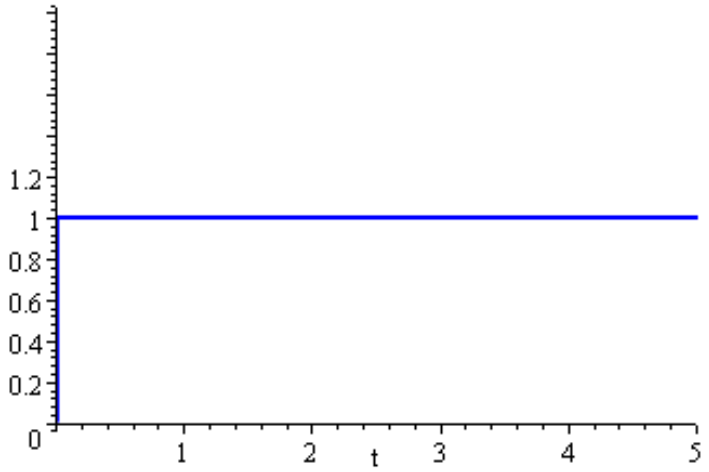
The *Transfer Function* provides a convenient way to find the response to inputs.

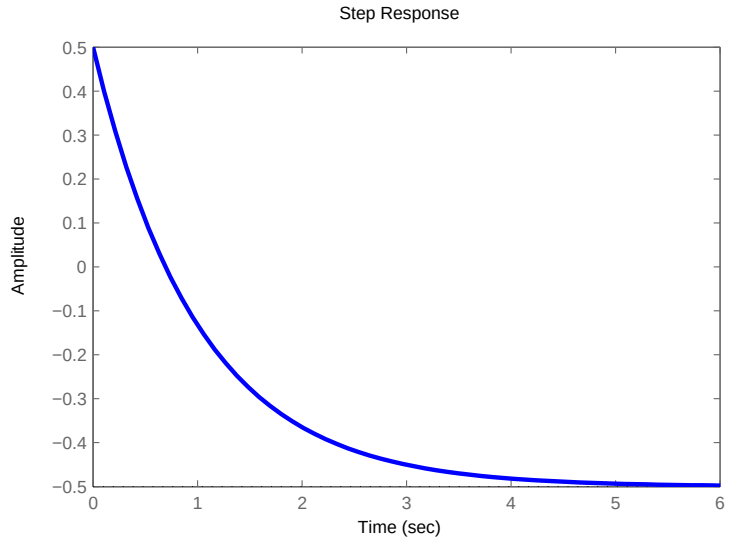
Step Input Response: $\hat{u}(s) = \frac{1}{s}$

$$\hat{y}(s) = \hat{G}(s)\hat{u}(s) = \frac{1}{2}\frac{s-1}{s+1}\frac{1}{s} = \frac{1}{2}\frac{s-1}{s^2+s}$$
$$= \frac{1}{2}\left(\frac{2}{s+1} - \frac{1}{s}\right)$$

Consulting our table of Laplace Transforms,

$$y(t) = \frac{1}{2}\mathcal{L}^{-1}\left(\frac{2}{s+1}\right) - \frac{1}{2}\mathcal{L}^{-1}\left(\frac{1}{s}\right)$$
$$= e^{-t} - \frac{1}{2}\mathbf{1}(t)$$





Using a Transfer Function to Find the Output

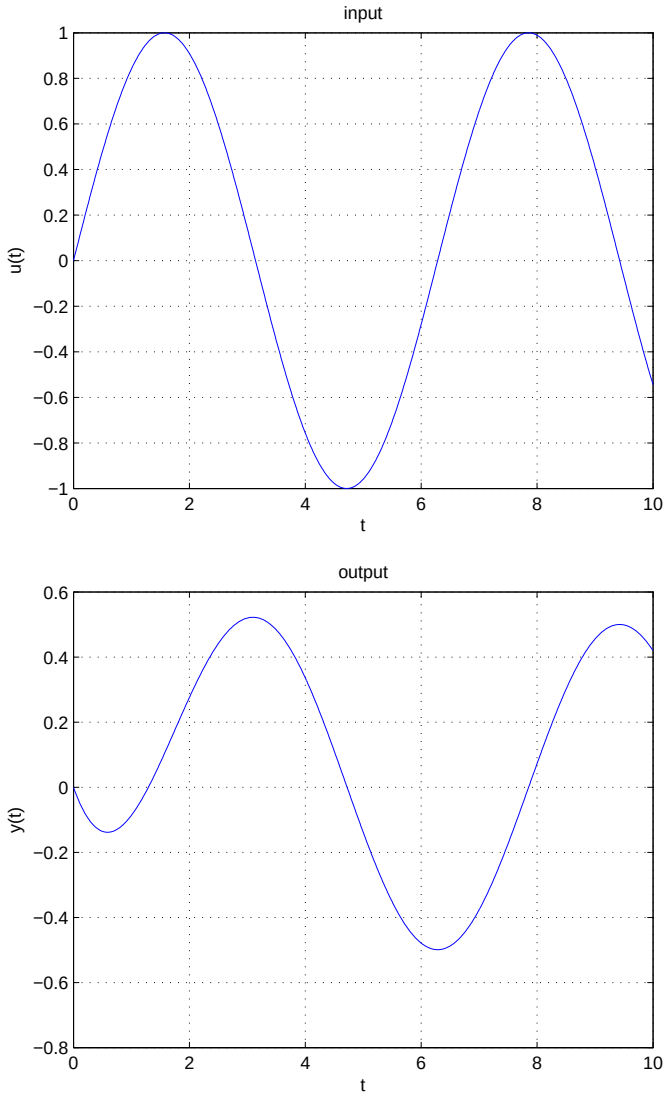
Example: Sinusoidal Input

Sine Function: $\hat{u}(s) = \frac{1}{s^2+1}$

$$\hat{y}(s) = \hat{G}(s)\hat{u}(s) = \frac{1}{2}\frac{s-1}{s+1}\frac{1}{s^2+1}$$
$$= \frac{1}{2}\frac{s-1}{s^3+s^2+s+1}$$
$$= \frac{1}{2}\left(\frac{s}{s^2+1} - \frac{1}{s+1}\right)$$

Consulting our table of Laplace Transforms,

$$y(t) = \frac{1}{2}\cos t - \frac{1}{2}e^{-t}$$



Note that this is the same answer we got by integration in Lecture 4.

Finding Transfer Functions from Dynamics

Example: Inverted Pendulum

Step 1: Write down the differential equations

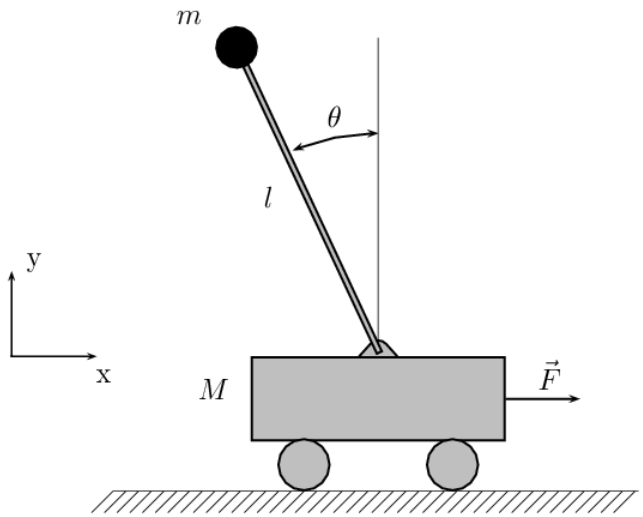
$$\ddot{\theta}(t) = \frac{Mgl}{2J}\theta(t) + \frac{1}{J}T(t) \qquad y(t) = \theta(t)$$

Step 2: Apply the Laplace transform to all equations:

$$s^2\hat{\theta}(s) = \frac{Mgl}{2J}\hat{\theta}(s) + \frac{1}{J}\hat{T}(s) \qquad \hat{y}(s) = \hat{\theta}(s)$$

Step 3: Solve for $\hat{\theta}(s)$ in terms of $\hat{T}(s)$

$$\hat{\theta}(s) = \frac{1}{J}\frac{1}{s^2 - \frac{Mgl}{2J}}\hat{T}(s)$$



Step 4: Solve for $\hat{y}(s)$ in terms of input $\hat{T}(s)$

$$\hat{y}(s) = \hat{\theta}(s) = \frac{1}{J}\frac{1}{s^2 - \frac{Mgl}{2J}}\hat{T}(s)$$

Step 5: Thus we have the Transfer Function:

$$\hat{G}(s) = \frac{\hat{y}(s)}{\hat{T}(s)} = \frac{1}{J}\frac{1}{s^2 - \frac{Mgl}{2J}}$$

Using the Transfer Function to Find the Output

Example: Impulse Response (Inverted Pendulum)

Impulse Input: $\hat{u}(s) = 1$

$$\hat{y}(s) = \hat{G}(s)\hat{u}(s) = \frac{1}{J}\frac{1}{s^2 - \frac{Mgl}{2J}}$$

$$= \frac{1}{J}\frac{1}{(s - \sqrt{\frac{Mgl}{2J}})(s + \sqrt{\frac{Mgl}{2J}})}$$

$$= \frac{1}{J}\sqrt{\frac{2J}{Mgl}}\left(\frac{1}{s - \sqrt{\frac{Mgl}{2J}}} - \frac{1}{s + \sqrt{\frac{Mgl}{2J}}}\right)$$

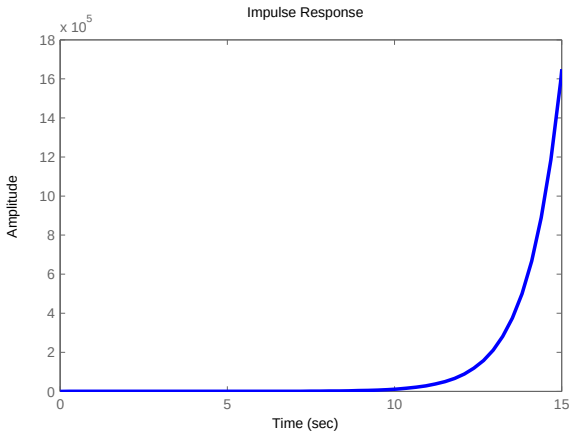


Figure: Impulse Response with $g = l = J = 1$, $M = 2$

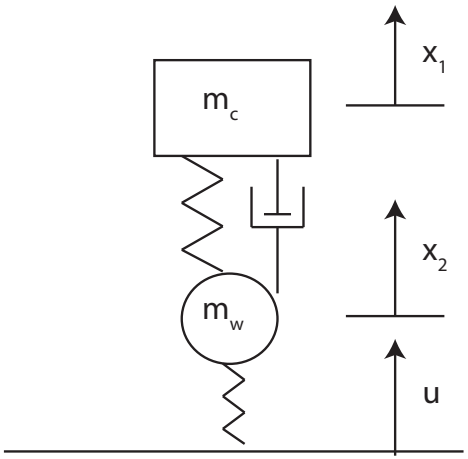
In time-domain:

$$y(t) = \frac{1}{J}\sqrt{\frac{2J}{Mgl}}\left(e^{\sqrt{\frac{Mgl}{2J}}t} - e^{-\sqrt{\frac{Mgl}{2J}}t}\right)$$

Conclusion: Inverted Pendulum is unstable (shock!)

Finding Transfer Functions from Dynamics

Example: Suspension System



Step 1: Write down the differential equations

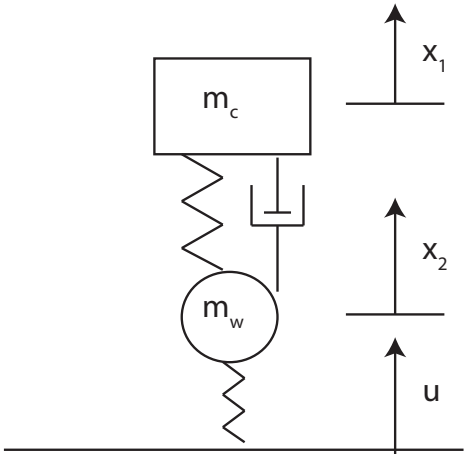
$$\ddot{z}_1(t) = -\frac{K_1}{m_c} z_1(t) - \frac{c}{m_c}\dot{z}_1(t) + \frac{K_1}{m_c} z_2(t) + \frac{c}{m_c}\dot{z}_2(t)$$

$$\ddot{z}_4(t) = \frac{K_1}{m_w} z_1(t) + \frac{c}{m_w}\dot{z}_1(t) - \left(\frac{K_1}{m_w} + \frac{K_2}{m_w}\right) z_2(t) - \frac{c}{m_w}\dot{z}_2(t) - \frac{K_2}{m_w} u(t)$$

$$y(t) = \begin{bmatrix} z_2(t) \end{bmatrix}$$

Finding Transfer Functions from Dynamics

Example: Suspension System



Step 2: Apply the Laplace transform to all equations:

$$s^2\hat{z}_1(s) = -\frac{K_1}{m_c}\hat{z}_1(s) - \frac{c}{m_c}s\hat{z}_1(s) + \frac{K_1}{m_c}\hat{z}_2(s) + \frac{c}{m_c}s\hat{z}_2(s)$$

$$s^2\hat{z}_2(s) = \frac{K_1}{m_w}\hat{z}_1(s) + \frac{c}{m_w}s\hat{z}_1(s) - \left(\frac{K_1}{m_w} + \frac{K_2}{m_w}\right)\hat{z}_2(s) - \frac{c}{m_w}s\hat{z}_2(s) - \frac{K_2}{m_w}\hat{u}(s)$$

$$\hat{y}(s) = \hat{z}_2(s)$$

Finding TFs from Dynamics: Suspension System

Step 3: Solve for $\hat{z}_1(s), \hat{z}_2(s)$ in terms of $\hat{u}(s)$

$$\left(s^2 + \frac{c}{m_c}s + \frac{K_1}{m_c}\right)\hat{z}_1(s) = \left(\frac{K_1}{m_c} + \frac{c}{m_c}s\right)\hat{z}_2(s)$$

$$\left(s^2 + \frac{c}{m_w}s + \frac{K_1}{m_w} + \frac{K_2}{m_w}\right)\hat{z}_2(s) = \left(\frac{K_1}{m_w} + \frac{c}{m_w}s\right)\hat{z}_1(s) - \frac{K_2}{m_w}\hat{u}(s)$$

First equation yields $\hat{z}_1(s) = \frac{\left(\frac{K_1}{m_c} + \frac{c}{m_c}s\right)}{\left(s^2 + \frac{c}{m_c}s + \frac{K_1}{m_c}\right)}\hat{z}_2(s)$. Now from second equation

$$\left(s^2 + \frac{c}{m_w}s + \frac{K_1}{m_w} + \frac{K_2}{m_w}\right)\hat{z}_2(s) = \frac{\left(\frac{K_1}{m_w} + \frac{c}{m_w}s\right)\left(\frac{K_1}{m_c} + \frac{c}{m_c}s\right)}{\left(s^2 + \frac{c}{m_c}s + \frac{K_1}{m_c}\right)}\hat{z}_2(s) - \frac{K_2}{m_w}\hat{u}(s)$$

$$\left(\left(s^2 + \frac{c}{m_w}s + \frac{K_1}{m_w} + \frac{K_2}{m_w}\right) - \frac{\left(\frac{K_1}{m_w} + \frac{c}{m_w}s\right)\left(\frac{K_1}{m_c} + \frac{c}{m_c}s\right)}{\left(s^2 + \frac{c}{m_c}s + \frac{K_1}{m_c}\right)}\right)\hat{z}_2(s) = -\frac{K_2}{m_w}\hat{u}(s)$$

$$\hat{z}_2(s) = -\left(\left(s^2 + \frac{c}{m_w}s + \frac{K_1}{m_w} + \frac{K_2}{m_w}\right) - \frac{\left(\frac{K_1}{m_w} + \frac{c}{m_w}s\right)\left(\frac{K_1}{m_c} + \frac{c}{m_c}s\right)}{\left(s^2 + \frac{c}{m_c}s + \frac{K_1}{m_c}\right)}\right)^{-1}\frac{K_2}{m_w}\hat{u}(s)$$

Finding TFs from Dynamics: Suspension System

Step 3: Solve for $\hat{z}_1(s), \hat{z}_2(s)$ in terms of $\hat{u}(s)$

$$\left(s^2 + \frac{c}{m_c}s + \frac{K_1}{m_c}\right)\hat{z}_1(s) = \left(\frac{K_1}{m_c} + \frac{c}{m_c}s\right)\hat{z}_2(s)$$

$$\left(s^2 + \frac{c}{m_w}s + \frac{K_1}{m_w} + \frac{K_2}{m_w}\right)\hat{z}_2(s) = \left(\frac{K_1}{m_w} + \frac{c}{m_w}s\right)\hat{z}_1(s) - \frac{K_2}{m_w}\hat{u}(s)$$

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$$\left(\left(s^2 + \frac{c}{m_w}s + \frac{K_1}{m_w} + \frac{K_2}{m_w}\right) - \frac{\left(\frac{K_1}{m_w} + \frac{c}{m_w}s\right)\left(\frac{K_1}{m_c} + \frac{c}{m_c}s\right)}{\left(s^2 + \frac{c}{m_c}s + \frac{K_1}{m_c}\right)}\right)\hat{z}_2(s) = -\frac{K_2}{m_w}\hat{u}(s)$$

$$\hat{z}_2(s) = -\left(\left(s^2 + \frac{c}{m_w}s + \frac{K_1}{m_w} + \frac{K_2}{m_w}\right) - \frac{\left(\frac{K_1}{m_w} + \frac{c}{m_w}s\right)\left(\frac{K_1}{m_c} + \frac{c}{m_c}s\right)}{\left(s^2 + \frac{c}{m_c}s + \frac{K_1}{m_c}\right)}\right)^{-1}\frac{K_2}{m_w}\hat{u}(s)$$

Now just use

$$\hat{z}_1(s) = \frac{\left(\frac{K_1}{m_c} + \frac{c}{m_c}s\right)}{\left(s^2 + \frac{c}{m_c}s + \frac{K_1}{m_c}\right)}\hat{z}_2(s)$$

To solve for $\hat{z}_1(s)$.

Finding Transfer Functions from Dynamics

Example: Suspension System

$$\hat{z}_2(s) = -\left(\left(s^2 + \frac{c}{m_w}s + \frac{K_1}{m_w} + \frac{K_2}{m_w}\right) - \frac{\left(\frac{K_1}{m_w} + \frac{c}{m_w}s\right)\left(\frac{K_1}{m_c} + \frac{c}{m_c}s\right)}{\left(s^2 + \frac{c}{m_c}s + \frac{K_1}{m_c}\right)}\right)^{-1} \frac{K_2}{m_w}\hat{u}(s)$$

Finding a common denominator and then inverting, we obtain:

$$\hat{z}_2(s) = \frac{K_2(m_c s^2 + cs + K_1)}{m_c m_w s^4 + c(m_w + m_c)s^3 + (K_1 m_c + K_1 m_w + K_2 m_c)s^2 + cK_2 s + K_1 K_2}\hat{u}(s)$$

Step 4: Solve for output $\hat{y}(s)$ in terms of input $\hat{u}(s)$

$$\hat{y}(s) = \hat{z}_2(s) = \frac{K_2(m_c s^2 + cs + K_1)}{m_c m_w s^4 + c(m_w + m_c)s^3 + (K_1 m_c + K_1 m_w + K_2 m_c)s^2 + cK_2 s + K_1 K_2}\hat{u}(s)$$

Step 5: Thus we have the Transfer Function:

$$\hat{G}(s) = \frac{\hat{y}(s)}{\hat{u}(s)} = \frac{K_2(m_c s^2 + cs + K_1)}{m_c m_w s^4 + c(m_w + m_c)s^3 + (K_1 m_c + K_1 m_w + K_2 m_c)s^2 + cK_2 s + K_1 K_2}$$

$$\hat{z}_2(s) = -\left(\left(s^2 + \frac{c}{m_w}s + \frac{K_1}{m_w} + \frac{K_2}{m_w}\right) - \frac{\left(\frac{K_1}{m_w} + \frac{c}{m_w}s\right)\left(\frac{K_1}{m_c} + \frac{c}{m_c}s\right)}{\left(s^2 + \frac{c}{m_c}s + \frac{K_1}{m_c}\right)}\right)^{-1} \frac{K_2}{m_w}\hat{u}(s)$$

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Compare to the State-Space Representation:

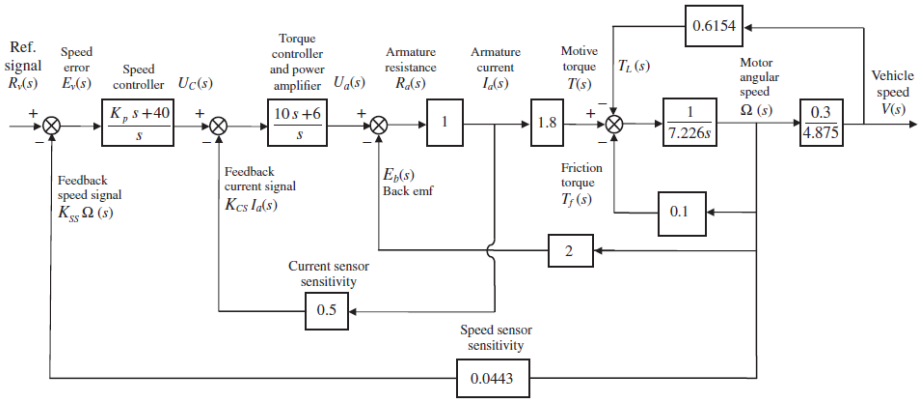
$$\frac{d}{dt}\begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix}(t) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{K_1}{m_c} & -\frac{c}{m_c} & \frac{K_1}{m_c} & \frac{c}{m_c} \\ 0 & 0 & 0 & 1 \\ \frac{K_1}{m_w} & \frac{c}{m_w} & -\left(\frac{K_1}{m_w} + \frac{K_2}{m_w}\right) & -\frac{c}{m_w} \end{bmatrix}\begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix}(t) + \begin{bmatrix} 0 \\ 0 \\ 0 \\ -\frac{K_2}{m_w} \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix}(t) + \begin{bmatrix} 0 \\ 0 \end{bmatrix} u(t)$$

Finding Transfer Functions from Block Diagram Algebra

Reduction by Familiar Forms

The interconnection of systems can be represent by block diagrams.



Our goal is to find the transfer function: $\hat{G}(s) = \dfrac{\hat{V}(s)}{\hat{R}_v(s)}$.

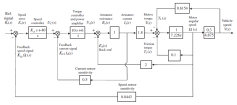
A **Brute Force Approach** is to write down all the equations for each block

- Need to label all the signals, $I_a(s)$, $T(s)$, $\Omega(s)$, et c.
- Write down one equation for each block and summing junction.
- Try to solve for $V(s)$.

However, before you use brute force, try Reducing as much of the block diagram as possible using **Familiar Forms:**

- Series (Cascade) Interconnections
- Parallel Interconnections
- Feedback Interconnections

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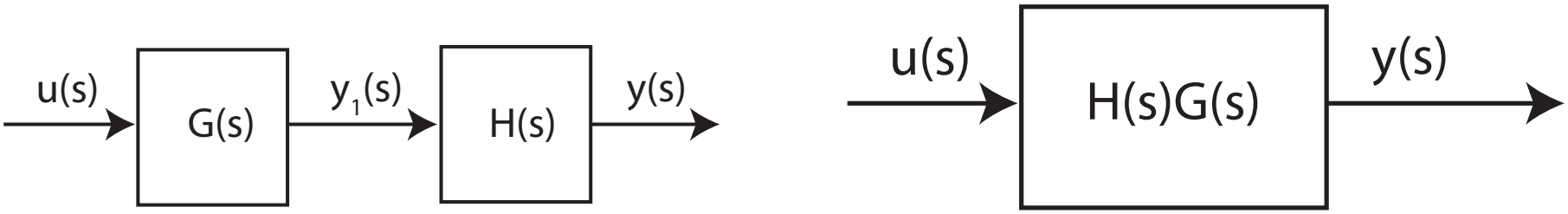
- Series (Cascade) Interconnections
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- Feedback Interconnections

The transfer function for this block diagram is

$$\hat{G}(s) = \frac{\hat{V}(s)}{\hat{R}_v(s)} = \frac{.1533(K_p s + 40)(s + .6)}{6s^3 + (3.613 + .1103K_p)s^2 + (4.47 + .06621K_p)s + 2.65}$$

Block Diagram Algebra : Reduction by Familiar Forms

Series Connection (Cascade)



For a **Series (or Cascade) Interconnection**:

- The output of G is the input to H.
- Let $\hat{G}(s)$ and $\hat{H}(s)$ be the transfer functions for G and H.
- Then

$$\hat{y}_1(s) = \hat{G}_1(s)\hat{u}(s) \qquad \hat{y}(s) = \hat{H}(s)\hat{y}_1(s) = \hat{H}(s)\hat{G}(s)\hat{u}(s)$$

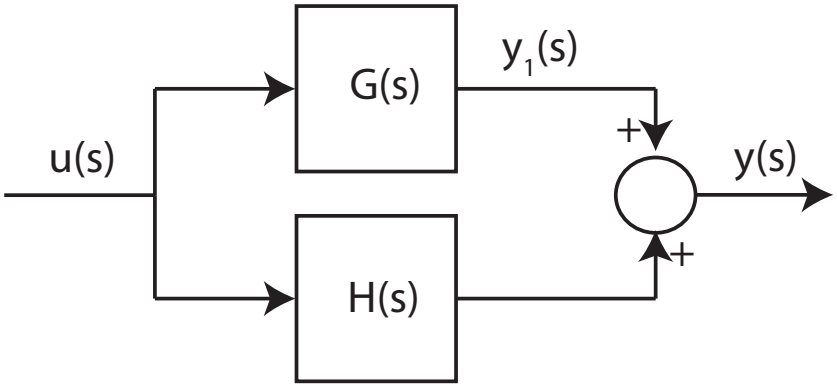
- The *Transfer Function*, $\hat{T}(s)$ for the combination of G and H is

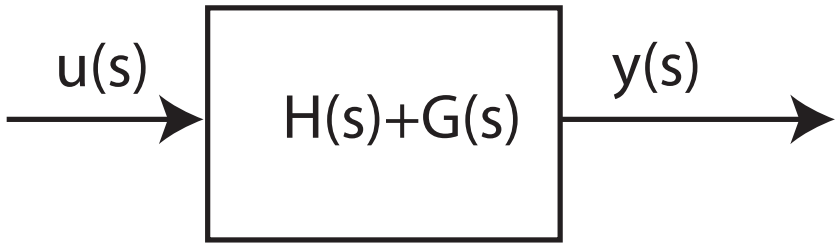
$$\hat{T}(s) = \frac{\hat{y}(s)}{\hat{u}(s)} = \hat{H}(s)\hat{G}(s)$$

Note: The order of the $\hat{G}$ and $\hat{H}$!

Block Diagram Algebra : Reduction by Familiar Forms

Parallel Connection





The Transfer function of a Parallel interconnection:

- Laplace transform:

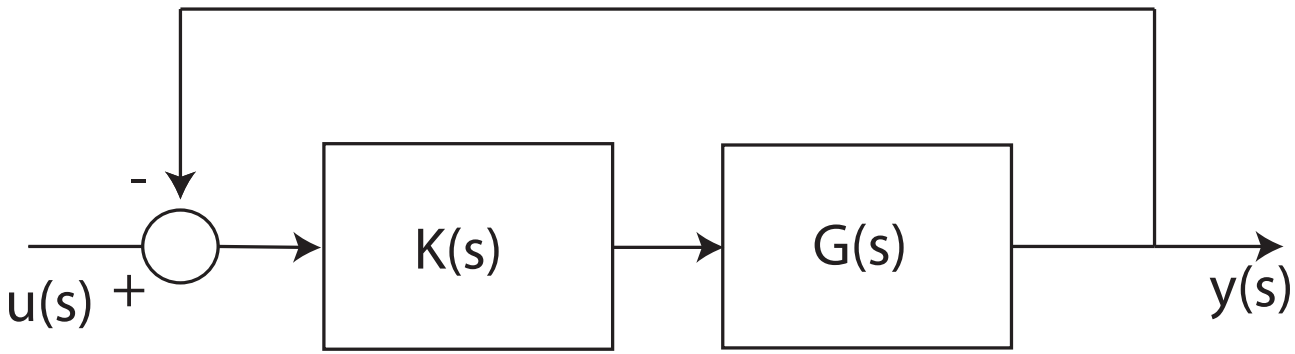
$$\hat{y}(s) = \hat{y}_1(s) + \hat{y}_2(s) = \hat{G}(s)\hat{u}(s) + \hat{H}(s)\hat{u}(s) = \left(\hat{H}(s) + \hat{G}(s)\right)\hat{u}(s)$$

- The *Transfer Function*, $\hat{T}(s)$ for the parallel interconnection of G and H is

$$\hat{T}(s) = \frac{\hat{y}(s)}{\hat{u}(s)} = \hat{H}(s) + \hat{G}(s)$$

Block Diagram Algebra : Reduction by Familiar Forms

Unitary Feedback Interconnection (Tracking)



Feedback:

- **Controller:** $z = K(u - y)$ **Plant:** $y = Gz$

In the Frequency Domain:

$$\hat{z}(s) = -\hat{K}(s)\hat{y}(s) + \hat{K}(s)\hat{u}(s) \qquad \hat{y}(s) = \hat{G}(s)\hat{z}(s)$$

so

$$\hat{y}(s) = \hat{G}(s)\hat{z}(s) = -\hat{G}(s)\hat{K}(s)\hat{y}(s) + \hat{G}(s)\hat{K}(s)\hat{u}(s)$$

Solving for $\hat{y}(s)$,

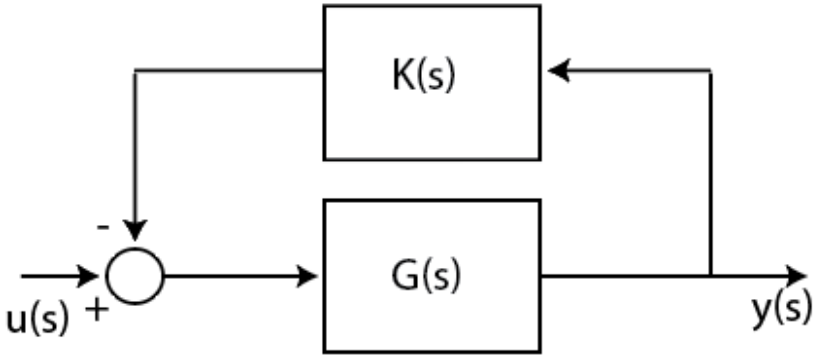
$$\hat{y}(s) = \frac{\hat{G}(s)\hat{K}(s)}{1 + \hat{G}(s)\hat{K}(s)}\hat{u}(s)$$

Block Diagram Algebra : Reduction by Familiar Forms

Controller Feedback Interconnection (Regulators)

When the Feedback has a block $(K(s))$

$$\hat{y}(s) = \hat{G}(s)\hat{z}(s)$$
$$\hat{z}(s) = u(s) - \hat{K}(s)\hat{y}(s)$$



Which yields

$$\hat{y}(s) = \hat{G}(s)\left(u(s) - \hat{K}(s)\hat{y}(s)\right) = \hat{G}(s)\hat{u}(s) - \hat{G}(s)\hat{K}(s)\hat{y}(s)$$

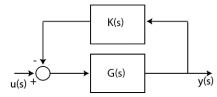
hence the Transfer Function is:

$$\hat{y}(s) = \frac{\hat{G}(s)}{1 + \hat{G}(s)\hat{K}(s)}\hat{u}(s).$$

Note: When $\hat{K} \to \infty$, $\hat{y} \to 0$.

When the Feedback has a block ($K(s)$)

$$\hat{y}(s) = \hat{G}(s)\hat{z}(s)$$
$$\hat{z}(s) = u(s) - \hat{K}(s)\hat{y}(s)$$



Which yields

$$\hat{y}(s) = \hat{G}(s)\left(u(s) - \hat{K}(s)\hat{y}(s)\right) = \hat{G}(s)\hat{u}(s) - \hat{G}(s)\hat{K}(s)\hat{y}(s)$$

hence the Transfer Function is:

$$\hat{y}(s) = \frac{\hat{G}(s)}{1 + \hat{G}(s)\hat{K}(s)}\hat{u}(s).$$

Note: When $\hat{K} \to \infty$, $\hat{y} \to 0$.

Pro Tip: If $\hat{G}(s) = \frac{n(s)}{d(s)}$ and $\hat{K}(s) = \frac{n_k(s)}{d_k(s)}$, then

Unitary Feedback:

$$\frac{\hat{G}(s)\hat{K}(s)}{1 + \hat{G}(s)\hat{K}(s)} = \frac{n(s)n_k(s)}{d(s)d_k(s) + n(s)n_k(s)}$$

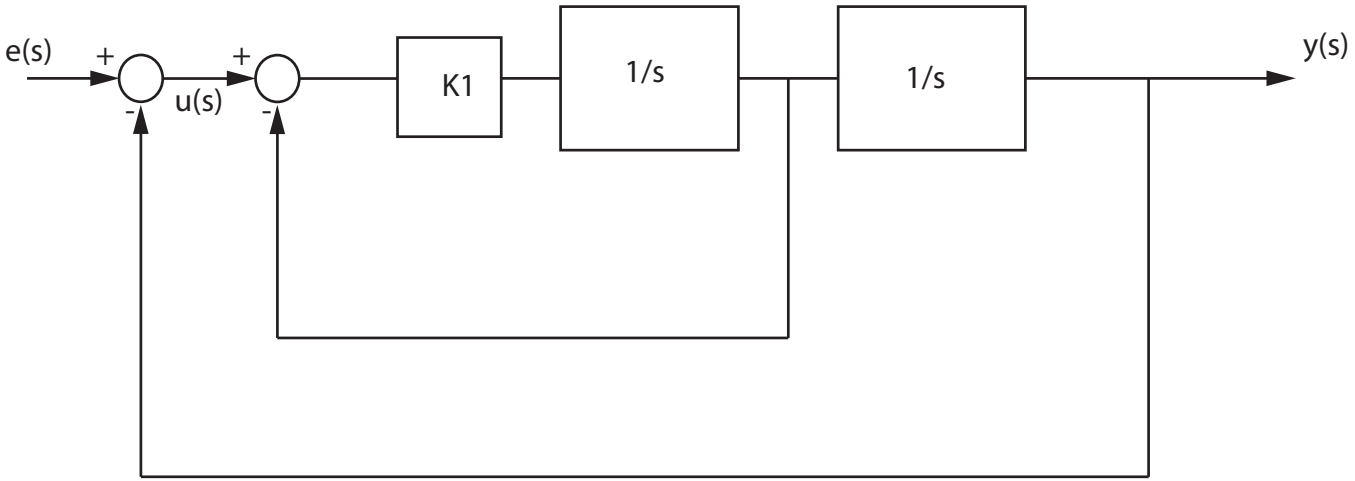
and **Controller Feedback:**

$$\frac{\hat{G}(s)}{1 + \hat{G}(s)\hat{K}(s)} = \frac{n(s)d_k(s)}{d(s)d_k(s) + n(s)n_k(s)}$$

Block Diagram Algebra : Reduction by Familiar Forms

Sequential Reduction of Feedback Loops

Now lets look at how to reduce a more complicated interconnections



Label

- The output from the inner loop z
- The input to the inner loop u

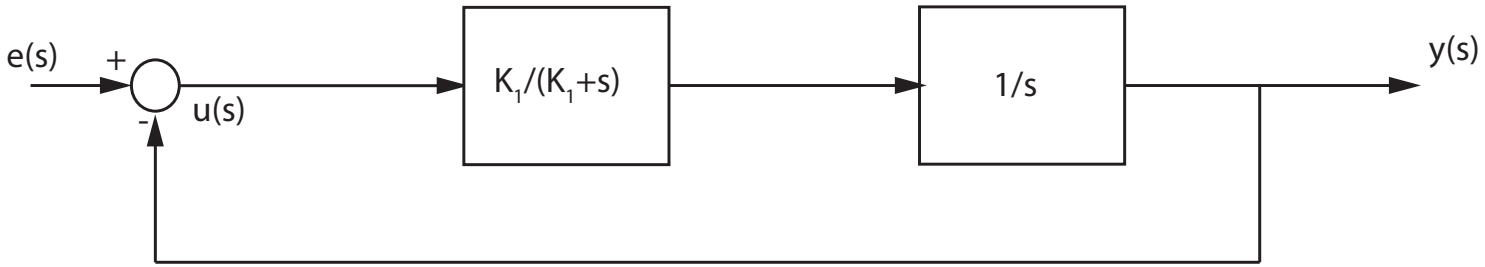
First **Close the Inner Loop** using the *Lower Feedback Interconnection*.

$$\hat{z}(s) = \frac{\frac{K_1}{s}}{\frac{K_1}{s}+1}\hat{u}(s) = \frac{K_1}{K_1+s}\hat{u}(s)$$

Block Diagram Algebra : Reduction by Familiar Forms

Sequential Reduction of Feedback Loops

We now have a reduced Block Diagram



Again, apply the *Lower Feedback Interconnetion*:

$$\hat{y}(s) = \frac{\frac{K_1}{s(K_1+s)}}{1+\frac{K_1}{s(K_1+s)}}\hat{e}(s) = \frac{K_1}{s(K_1+s)+K_1}\hat{e}(s)$$

So the Transfer function is $\hat{T}(s) = \dfrac{\hat{y}(s)}{\hat{e}(s)} = \dfrac{K_1}{s^2+K_1 s+K_1}$

Block Diagram Algebra : Reduction by Familiar Forms

Illustration

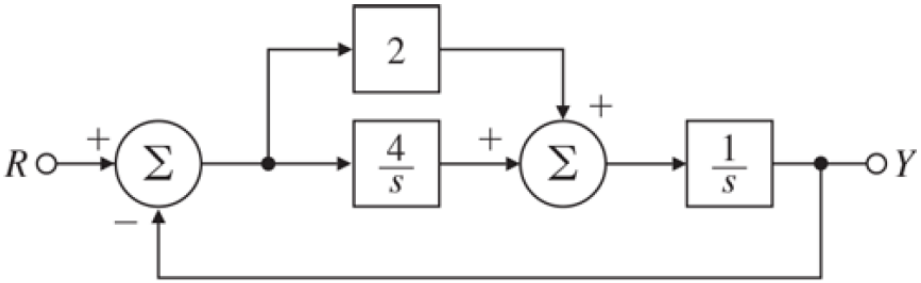


Figure: First reduce parallel interconnection

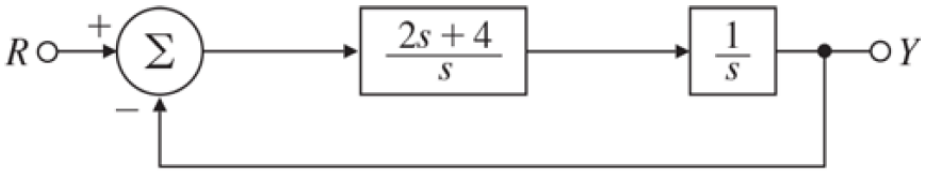


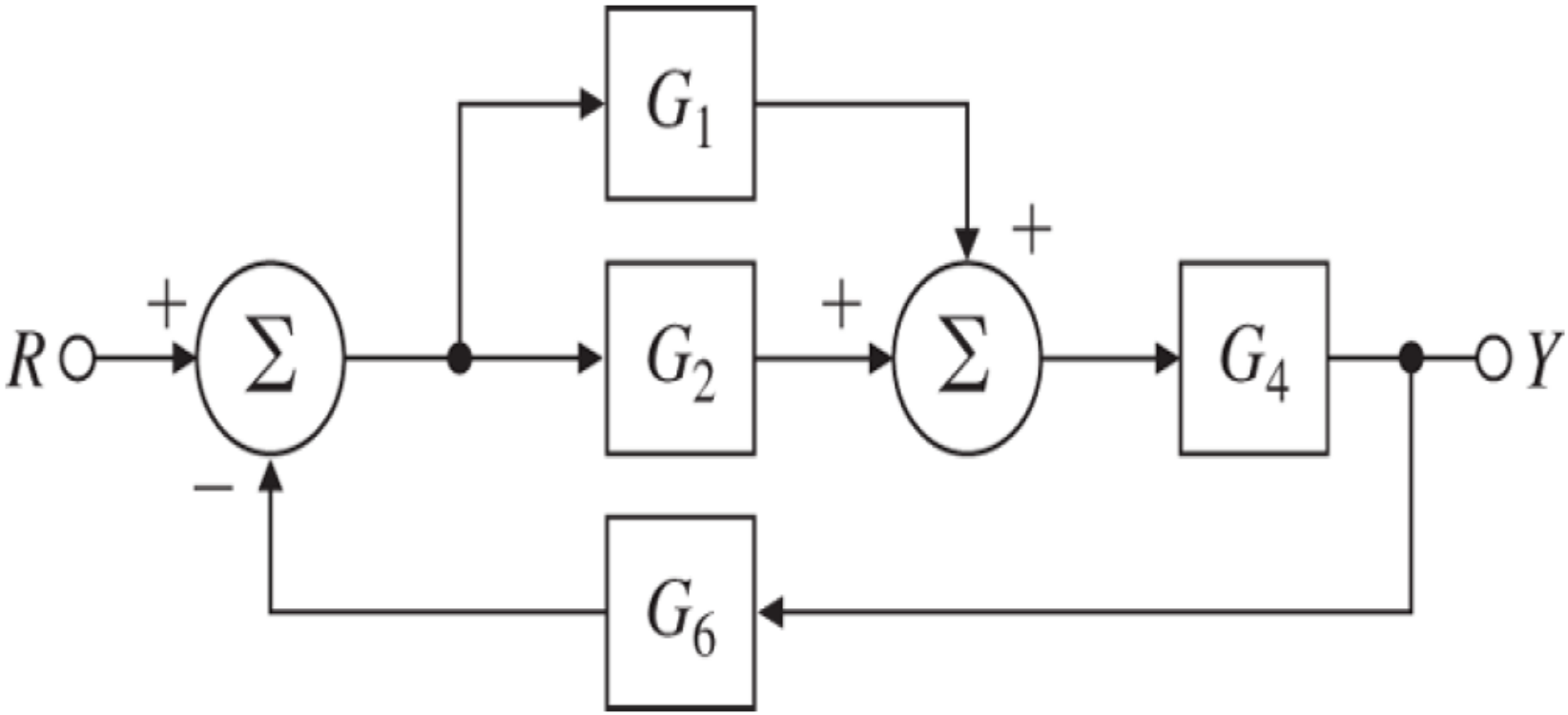
Figure: Then close the loop

Final :

$$G(s) = \frac{GK}{1+GK} = \frac{\frac{2s+4}{s^2}}{1+\frac{2s+4}{s^2}} = \frac{2s+4}{s^2+2s+4}$$

Block Diagram Algebra : Reduction by Familiar Forms

Illustration



Block Diagram Algebra : Reduction by Familiar Forms

Illustration

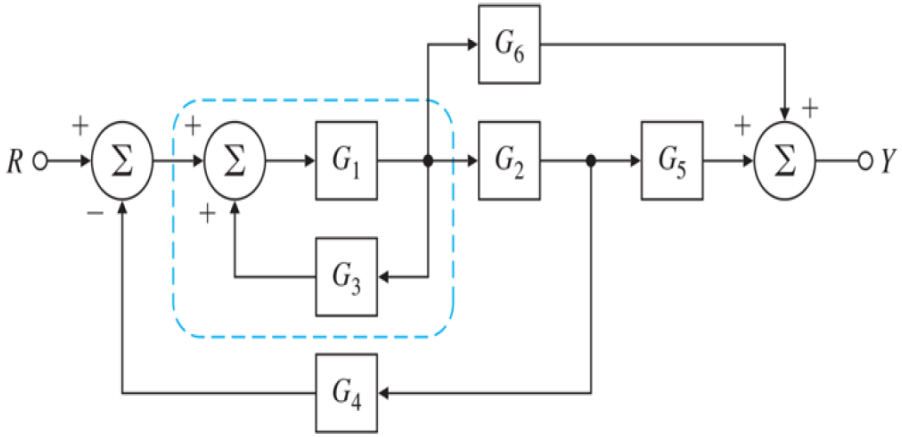


Figure: Identify/reduce first feedback loop

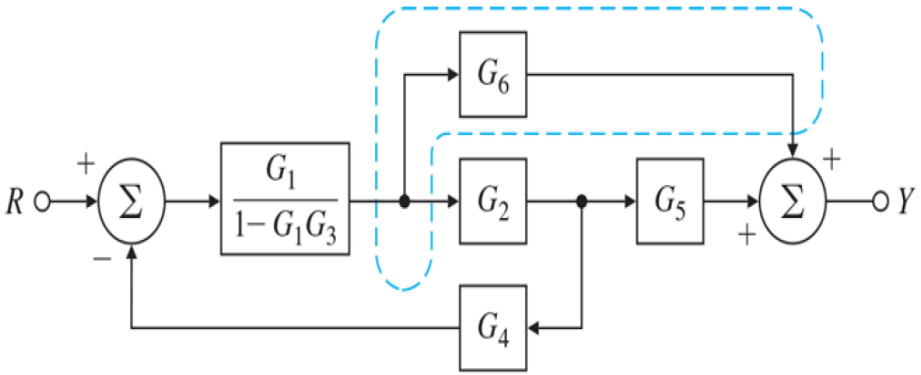
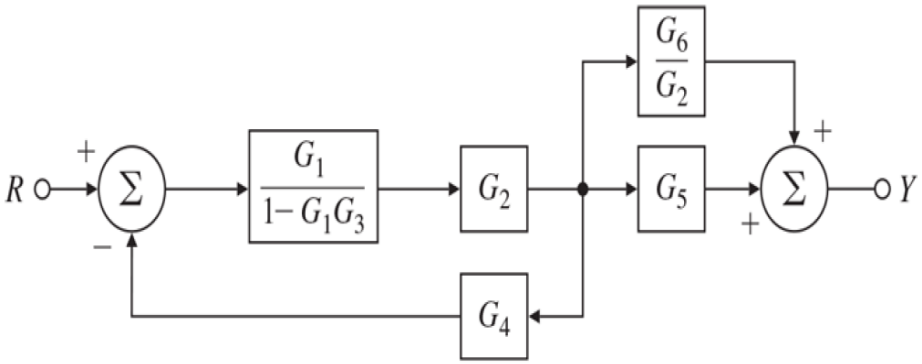


Figure: Identify parallel interconnection (move pickup point)



Loop:

$$\frac{\frac{G_1G_2}{1-G_1G_3}}{1+\frac{G_1G_2}{1-G_1G_3}G_4} = \frac{G_1G_2}{1-G_1G_3+G_1G_2G_4}$$

Parallel: $\frac{G_6}{G_2} + G_5 = \frac{G_6+G_5G_2}{G_2}$

Final (Series):

$$G(s) = \frac{G_6+G_5G_2}{G_2}\frac{G_1G_2}{1-G_1G_3+G_1G_2G_4} = \frac{G_1(G_6+G_5G_2)}{1-G_1G_3+G_1G_2G_4}$$

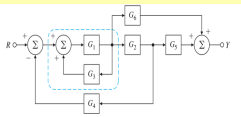


Figure: Identify/reduce first feedback loop

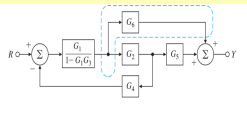
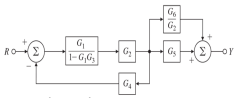


Figure: Identify parallel interconnection (move pickup point)

Loop:

$$\frac{\frac{G_1G_2}{1-G_1G_3}}{1+\frac{G_1G_2}{1-G_1G_3}G_4} = \frac{G_1G_2}{1-G_1G_3+G_1G_2G_4}$$



Parallel: $\frac{G_6}{G_2} + G_5 = \frac{G_6+G_5G_2}{G_2}$

Final (Series):

$$G(s) = \frac{G_6+G_5G_2}{G_2}\,\frac{G_1G_2}{1-G_1G_3+G_1G_2G_4} = \frac{G_1(G_6+G_5G_2)}{1-G_1G_3+G_1G_2G_4}$$

Notice how we moved the pickup point!

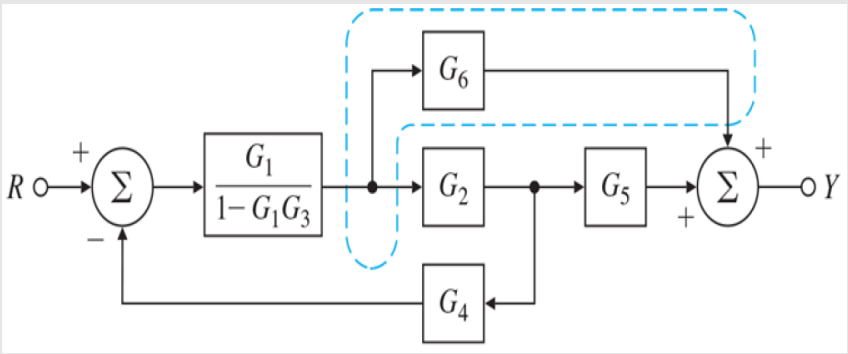


Figure: Initial signal **inside** feedback loop

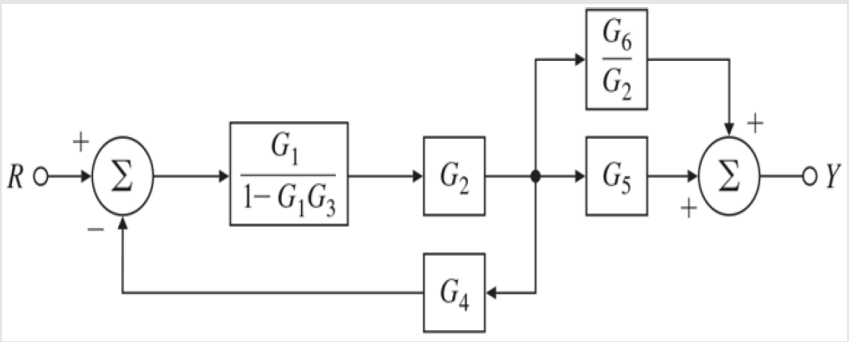


Figure: Need to move it **outside** before reducing the loop

We created a new signal $\hat{x} \to \frac{1}{G_2}\hat{x}$ and fed that to G_6 instead of $\hat{x}$!

The Effect of Feedback: Impulse Response

Inverted Pendulum Model

Transfer Function

$$\hat{G}(s) = \frac{1}{Js^2 - \frac{Mgl}{2}}$$

Controller: Static Gain: $\hat{K}(s) = K$
Input: Impulse: $\hat{u}(s) = 1$.
Closed Loop: Lower Feedback

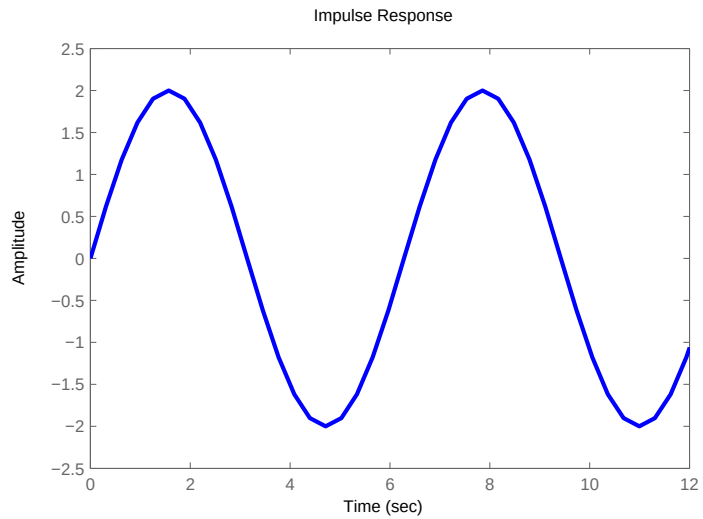
$$\hat{y}(s) = \frac{\hat{G}(s)\hat{K}(s)}{1 + \hat{G}(s)\hat{K}(s)}\hat{u}(s) = \frac{\frac{K}{Js^2 - \frac{Mgl}{2}}}{1 + \frac{K}{Js^2 - \frac{Mgl}{2}}} = \frac{K}{Js^2 - \frac{Mgl}{2} + K}$$

First Case:

- If $K > \frac{Mgl}{2}$, then $K - \frac{Mgl}{2} > 0$, so

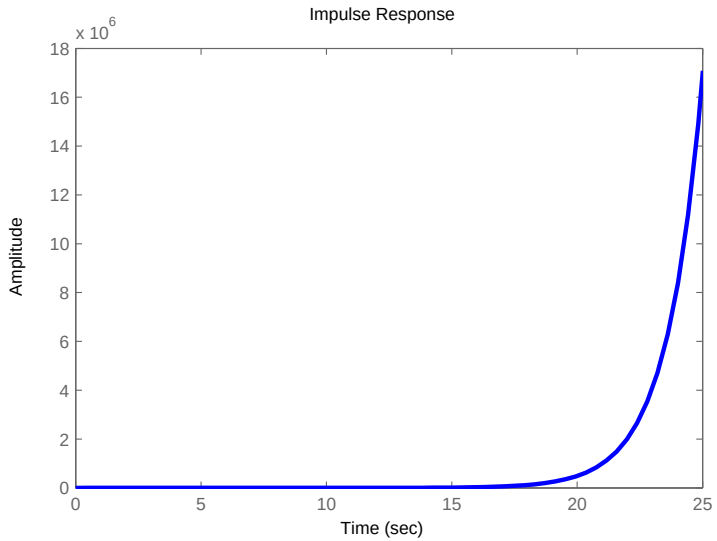
$$\hat{y}(s) = \frac{K/J}{s^2 + \left(K/J - \frac{Mgl}{2J}\right)}$$

$$y(t) = \frac{K}{J\sqrt{K/J - \frac{Mgl}{2J}}} \sin\left(\sqrt{K/J - \frac{Mgl}{2J}}t\right)$$



The Effect of Feedback: Impulse Response

Inverted Pendulum Model



Second Case:

- If $K < \frac{Mgl}{2}$, then $K - \frac{Mgl}{2} < 0$, so

$$\hat{y}(s) = \frac{K}{J}\left(\frac{1}{s - \sqrt{K/J - \frac{Mgl}{2J}}} + \frac{1}{s + \sqrt{K/J - \frac{Mgl}{2J}}}\right)$$

$$y(t) = \frac{K}{J}\left(e^{\sqrt{K/J - \frac{Mgl}{2J}}t} + e^{-\sqrt{K/J - \frac{Mgl}{2J}}t}\right)$$

Important: Value of K determines stability vs. instability

Summary

What have we learned today?

Transfer Functions

- Transfer Function Representation of a System
- State-Space to Transfer Function
- Direct Calculation of Transfer Functions

Block Diagram Algebra

- Modeling in the Frequency Domain
- Reducing Block Diagrams

Next Lecture: Partial Fraction Expansion